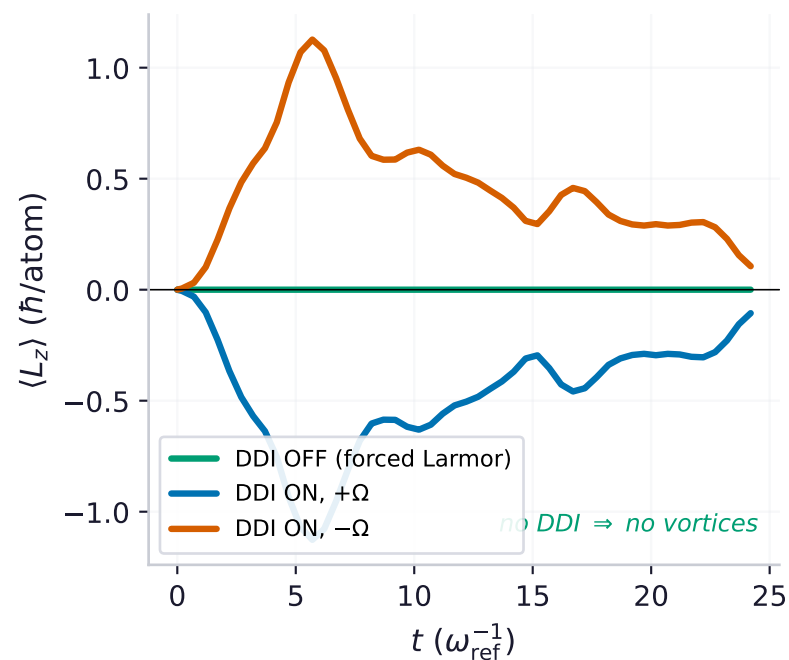
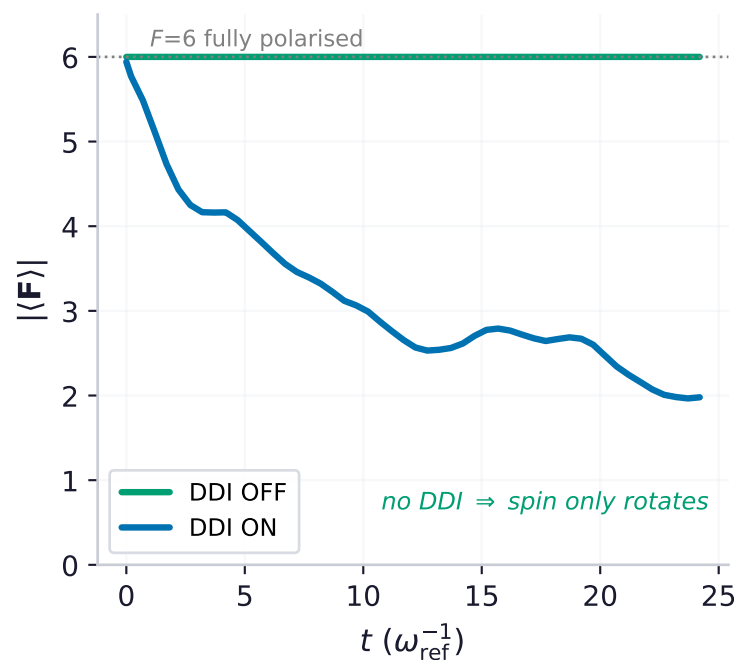


Rotating magnetic field $\rightarrow$ quantum vortices (Einstein-de Haas) + Barnett spin, controlled by rotation direction
¹⁵¹Eu $F=6$ dipolar BEC — vortices REQUIRE the DDI (zero without it); the $\langle F_z \rangle$ swing alone is mostly forced Larmor precession

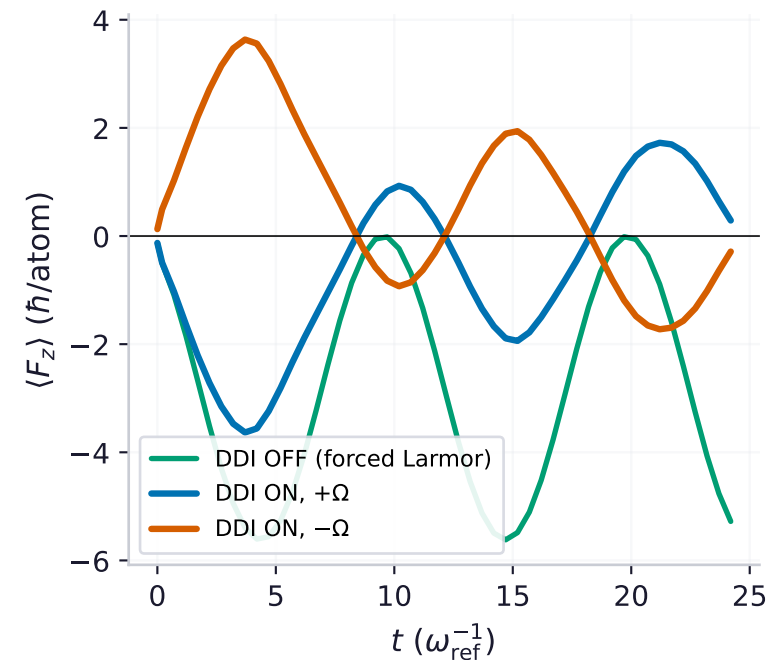
(a) vortices $\langle L_z \rangle$ — need DDI



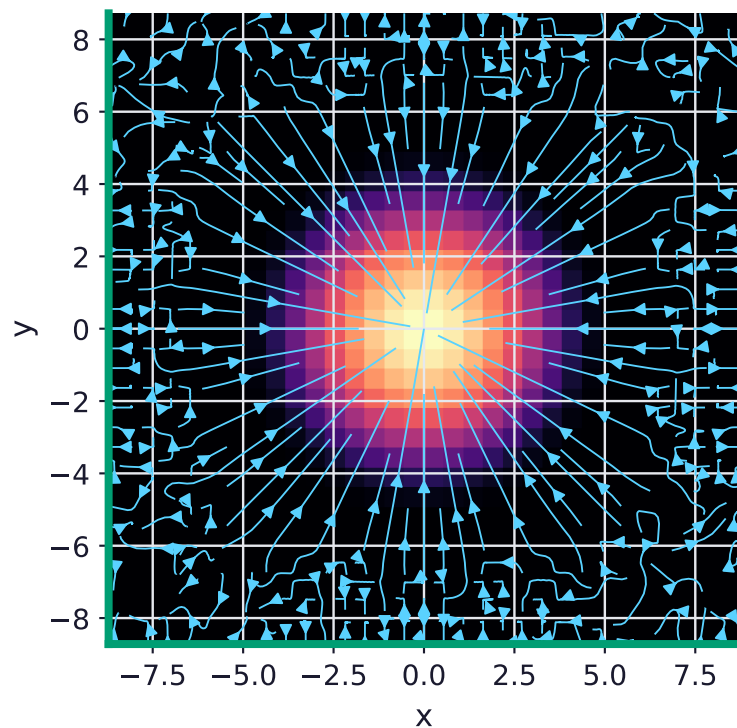
(b) spin length $|F|$ — DDI depolarises



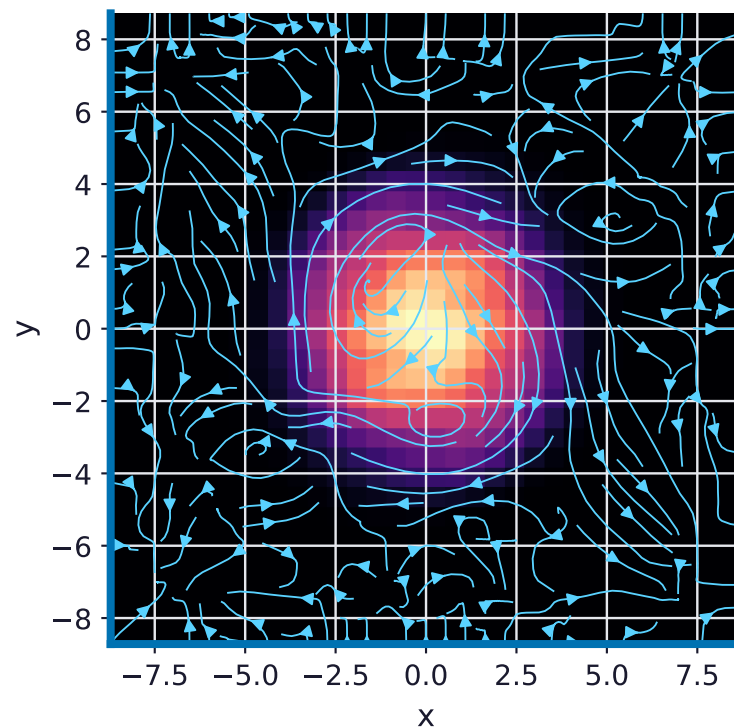
(c) $\langle F_z \rangle$ — Larmor + Barnett



(d) DDI OFF — no vortex



(e) DDI ON, $+\Omega$ — vortex (CCW)



(f) DDI ON, $-\Omega$ — vortex (CW)

